MIL-P-60086 (MU)
3 November 1964

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M62
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M62.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packing and Marking for Interplant Shipment of Inert Ammunition Components; General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1390

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DRAWINGS

ORDNANCE CORPS

8838012 - Cup, Battery
8838015 - Plug, Firing
8838085 - Head
8838089 - Body

PUBLICATIONS

ORDNANCE CORPS

IEL-8780970 - Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwgs.) 8838012, 8838015, 8838085 and 8838089, as applicable, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The tubing to be fabricated into primer bodies shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks and without permanent deformation beyond drawing dimensions when tested as specified in 4.4.1 (see 6.4).

3.4 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

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4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor inspection system.-The contractor shall provide and maintain an effective inspection system in compliance with the requirements of Specification MIL-I-45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.6. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.
b. Number and revision date of specification.
c. Characteristic code number.
d. Lot or serial number.
e. Number of defectives found.
f. Assignable cause or analysis of occurrence.
g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 At the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.5) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of (25) twenty-five complete set of parts (i.e., (25) twenty-five each of every

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component which have been produced by the contractor or furnished by a supplier and which has been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of inert components

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produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235, when applicable).

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, battery (see dwg. 8838012).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Major:	AQL 0.65 percent		
101.	Outside diameter, min	Gage	01001
102.	Thickness through bottom, maximum (max)	Gage	01002
103.	Length	Gage	01003
Minor:	AQL 2.50 percent		
201.	Diameter of hole	Gage	01004
202.	Thickness through bottom. min	Gage	01005
203.	Inside diameter	Gage	01006
204.	Outside diameter, max	Gage	01007
205.	Flat missing from surface at open end	Visual	01008
206.	Evidence of poor workmanship (see 3.4)	Visual	01009

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4.3.2.2 Plug, firing (see dwg. 8838015).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Large diameter, min.	Gage	02001
102.	Length and profile of point	Gage	02002
103.	Length of pilot, min.	Gage	02003
104.	Total length	Gage	02004
Minor:	AQL 2.5 percent		
201.	Large diameter, max.	Gage	02005
202.	Length of pilot, max.	Gage	02006
203.	Diameter of pilot	Gage	02007
204.	Finish improper	Visual	02008
205.	Evidence of poor workmanship (see 3.4)	Visual	02009

4.3.2.3 Head (see dwg. 8838085).

Categories	Defects	Method Of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.0 percent		
101.	Pitch diameter of body thread	Gage	03001
102.	Major diameter of body thread, min.	Gage	03002
103.	Pitch diameter of case thread	Gage	03003
104.	Major diameter of case thread	Gage	03004
105.	Diameter of body pilot, max.	Gage	03005
106.	Diameter of battery cup cavity	Gage	03006
107.	Diameter of percussion element cavity	Gage	03007

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108.	Depth of percussion element cavity	Gage	03008
109.	Thickness of flange, max	Gage	03009
110.	Crack or split	Visual	03010
111.	Flash hole missing or obstructed	Visual	03011

Minor: AQL 6.5 percent

201.	Major diameter of body thread, max	Gage	03012
202.	Diameter of flange	Gage	03013
203.	Depth of battery cup cavity,	Gage	03014
204.	Diameter of body pilot, min	Gage	03015
205.	Concentricity between percussion element cavity and flash hole	Gage	03016
206.	Concentricity of percussion element cavity with case thread	Gage	03017
207.	Concentricity of flange with case thread	Gage	03018
208.	Diameter of chamfer at mouth of percussion element cavity	Gage	03019
209.	Diameter of flash hole	Gage	03020
210.	Length to shoulder above body thread	Gage	03021
211.	Length of body thread undercut, max	Gage	03022
212.	Length of case thread undercut, max	Gage	03023
213.	Length from rear face to case thread shoulder	Gage	03024
214.	Diameter of shoulder between threads	Gage	03025
215.	Distance between wrench slots	Gage	03026
216.	Depth of wrench slots, min	Gage	03027
217.	Total length	Gage	03028
218.	Depth of flash cavity	Gage	03029

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219.	Diameter and profile of flash cavity	Gage	03030
220.	Length of body thread including thread undercut	Gage	03031
221.	Thread undercut missing	Visual	03032
222.	Radius missing	Visual	03033
223.	Finish improper	Visual	03034
224.	Evidence of poor workmanship (see 3.4)	Visual	03035

4.3.2.4 Body (see dwg. 8838089).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Flash hole missing	Gage	04001*
2.	Crack or split	Visual	04002*
Major:	AQL 1.0 percent		
101.	Pitch diameter of thread	Gage	04003
102.	Length of thread, min	Gage	04004
103.	Minor diameter of thread, max	Gage	04005
104.	Wall thickness, min	Gage	04006
105.	Thread stripped or otherwise damaged	Visual	04007
Minor:	AQL 4.0 percent		
201.	Length of thread, max	Gage	04008
202.	Location of first row of holes	Gage	04009
		Gage	04010
203.	Inside diameter, min	Gage	04011
204.	Depth of counterbore, max	Gage	04012
205.	Diameter of counterbore	Gage	04013
206.	Total length, max	Gage	04014
207.	Plug gage fails to pass completely through body	Gage	04015
208.	Body fails to pass through required ring gage		

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209.	Concentricity of inside diameter with pitch diameter of thread	Gage	04016
210.	Finish improper	Visual	04017
211.	Protective coating damaged (base metal exposed)	Visual	04018
212.	Evidence of poor workmanship (see 3.4)	Visual	04019

4.3.3. Testing

4.3.3.1 Hydrostatic test (see 3.3) Critical defect - Code No. 05001*.-This test shall be conducted 100 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 8780970 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for

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determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures

4.4.1 Hydrostatic test.-Prior to machining, the tubing for primer fabrication shall be assembled into the test equipment specified in 4.4.3.4 and shall be tested for not less than five seconds at not less than the hydrostatic pressure calculated from the following formula:

P = (2ST times .9) divided by D.

where:

P = test pressure in pounds per square inch (p.s.i)
S = min. specified yield strength, p.s.i.
T = specified min. wall thickness in inches.
D = specified max. outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Preservation and packaging of parts for Primer, Percussion, M62, shall be in accordance with Military Specification MIL-P-10025.

5.2 Marking.-Marking shall be in accordance with Military Specification MIL-P-10025.

5.3 Data cards.-Data card information shall be as specified in Military Specification MIL-A-2550.

6. NOTES

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6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary

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to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Hydrostatic test.-If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, and has successfully met the requirements, a second hydrostatic test will not be required.

6.5 Submission of initial production, Pre-production or Qualification samples.-Instructions, as to the location for the evaluation of Initial Production, Pre-production, or product qualification shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly.

6.6 Critical defect report

6.6.1 Submission of critical defect reports.-For U. S. Army Munitions Command procurements, one copy of each critical defect

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report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUPA-RO; Picatinny Arsenal, ATTN: SMUPA-ND.

6.6.2 Reporting to other services.-For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1390-A418

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DEPARTMENT OF THE ARMY

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DEPARTMENT OF THE ARMY

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DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

COMMANDING OFFICER
UNITED STATES ARMY
PICATINNY ARSENAL
ATTN: SMUPA-ND
DOVER, NEW JERSEY 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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